PAGE REPLACEMENT ALGORITHM SIMULATOR

PROJECT REPORT

Operating Systems Lab | Virtual Memory and Page Replacement

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Programming Language	Python 3
Notebook	Jupyter Notebook (.ipynb)
Algorithms	FIFO, LRU, and Optimal
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Abstract

This project simulates three page replacement policies used in operating systems: First-In, First-Out (FIFO), Least Recently Used (LRU), and Optimal. The program accepts a page-reference string and the number of available frames, then displays a step-by-step trace showing frame contents, page hits, page faults, and replaced pages. It also compares the algorithms using page-fault counts and hit/fault ratios.

1. Introduction

In a demand-paged system, only a subset of a process's pages is kept in main memory. When a requested page is not present in memory, a page fault occurs. If all frames are occupied, the operating system must select a page to evict before loading the requested page. A page replacement policy defines how that victim page is selected.

2. Objectives

• Implement FIFO, LRU, and Optimal page replacement algorithms.
• Accept a page-reference string and frame count at runtime.
• Validate numeric input and reject negative page numbers or invalid frame counts.
• Display a step-by-step trace of frame contents, hits, faults, and replaced pages.
• Compare the algorithms using page-fault totals and hit/fault ratios.

3. Algorithms

Algorithm	Victim selection rule	Key property
FIFO	Replace the page that entered memory earliest.	Simple to implement; may exhibit Belady's anomaly.
LRU	Replace the page that has not been used for the longest time in the past.	Uses recent-use history as a practical heuristic.
Optimal	Replace the page whose next use is farthest in the future, or which is never used again.	Benchmark for minimum possible faults when future references are known.

4. Methodology

The simulator parses comma- or space-separated page numbers and validates the frame count. Each algorithm processes the reference string from left to right. For every page reference, it checks whether the page is already resident (a hit) or must be loaded (a fault). When memory is full, the algorithm's victim-selection rule determines which page is replaced. A trace row is recorded for every step.

5. Implementation

The source code is organized into input parsing and validation functions, three independent algorithm functions (FIFO, LRU, and Optimal), a trace-printing function, a comparison function, and a main entry point. The standalone Python source can run in a terminal with Python 3 and does not require pandas or notebook-specific display functions.

6. Sample Execution and Results

For the sample reference string **7 0 1 2 0 3 0 4 2 3 0 3 2** with **3 frames**, the tested implementation produces the following totals:

Algorithm	Frames	Page Faults	Page Hits	Hit Ratio	Fault Ratio
FIFO	3	10	3	23.08%	76.92%
LRU	3	9	4	30.77%	69.23%
Optimal	3	7	6	46.15%	53.85%

These values are a sample test case for this report, not a claim that every possible reference string produces the same ranking.

7. Belady's Anomaly

Belady's anomaly is the phenomenon in which increasing the number of frames can increase the number of page faults for FIFO on certain reference strings. The notebook experiments associated with this project investigate FIFO results at different frame counts. Whether the anomaly occurs depends on the selected reference string.

8. Conclusion

The project demonstrates how FIFO, LRU, and Optimal make different page replacement decisions for the same reference string. The step-by-step traces make each decision visible, while the summary comparison quantifies page faults and hits. Optimal provides a useful theoretical benchmark; FIFO is simple, and LRU uses past access history.

9. How to Run

1. Install Python 3.
2. Open a terminal in the folder containing `page_replacement_simulator.py`.
3. Run `python3 page_replacement_simulator.py`.
4. Enter the page-reference string and number of frames when prompted.

Project files: page_replacement_simulator.py, Jupyter Notebook (.ipynb), README.md, and execution screenshots.